

Unit 4 - The Coordinate Plane

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The following B.E.S.T. Standards will be covered in this unit:

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x – or y - axis as the line of reflection when two ordered pairs have an opposite x – or y –coordinate.

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x –coordinate or the same y –coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.

- *Clarification 1: Instruction includes finding distances between points, computing dimensions of a rectangle or determining a fourth vertex of a rectangle.*
- *Clarification 2: Problems involving rectangles are limited to cases where the sides are parallel to the axes.*

Unit 4, Lesson 1: Plotting Integers Using One-Unit Scales for Axes



Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.



Learning Target

- ☐ I can plot with integers using one-unit scales for axes.



4.1.1 Warm-Up: Guess the Number

An image of $\frac{1}{4}$ cup of candy corn is shown.



- Estimate the number of pieces of candy corn that are in the measuring cup.
- Ask two classmates the value they estimated. Record their value and write your summary of their reason. You are the only person who should write in the first two columns of the table. Have the person initial next to your summary stating that your summary is correct.

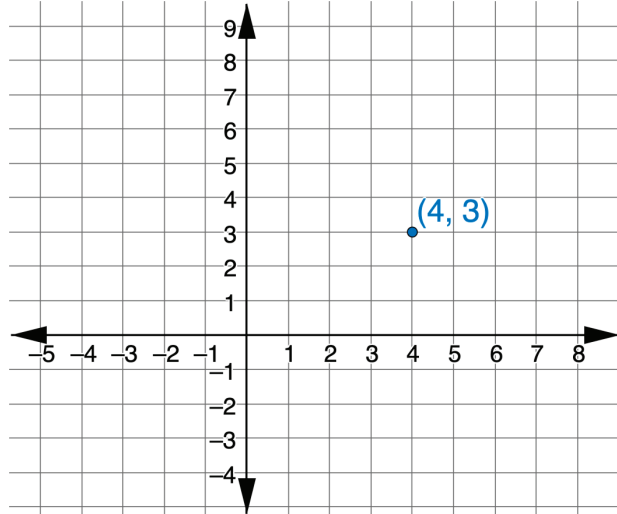
Guess	My Summary of Their Reason	Initials

- How close was your estimate to the actual number?



4.1.2 Exploration: The Coordinate Plane

1. This graph shows a point that is plotted on a coordinate plane.



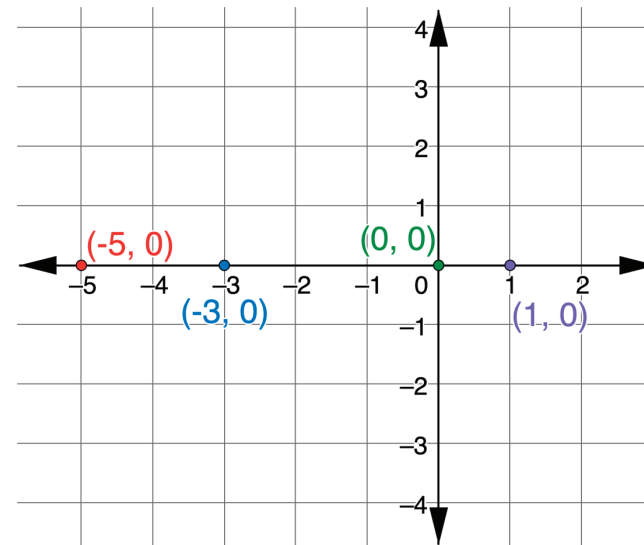
The two numbers are called the coordinates. The two numbers are also known as an ordered pair, (x, y) .

- a. Discuss with your partner which coordinate will change if the point is dragged from left to right.
- b. Discuss with your partner which coordinate will change if the point is dragged up or down.

- c. If the point $(4, 3)$ could be dragged around on the coordinate grid, what changes would you expect to see in the ordered pair? Select all that apply.

- ☐ The x -coordinate becomes smaller when the point is moved to the left.
- ☐ The x -coordinate becomes larger when the point is moved to the left.
- ☐ The x -coordinate stays the same when the point is moved to the left.
- ☐ The y -coordinate becomes smaller when the point is moved up.
- ☐ The y -coordinate becomes smaller when the point is moved down.
- ☐ The y -coordinate stays the same when the point is moved up.

2. Four points are plotted on the coordinate grid.



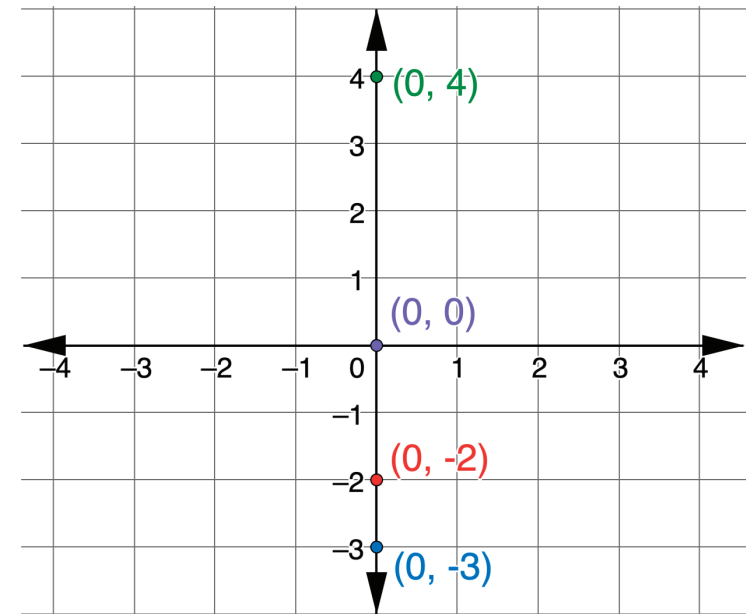
- a. What do you notice about all four points?

b. What is the same about the points?

c. What is different about the points?

d. Discuss with your partner what is true about all the points on the x -axis.

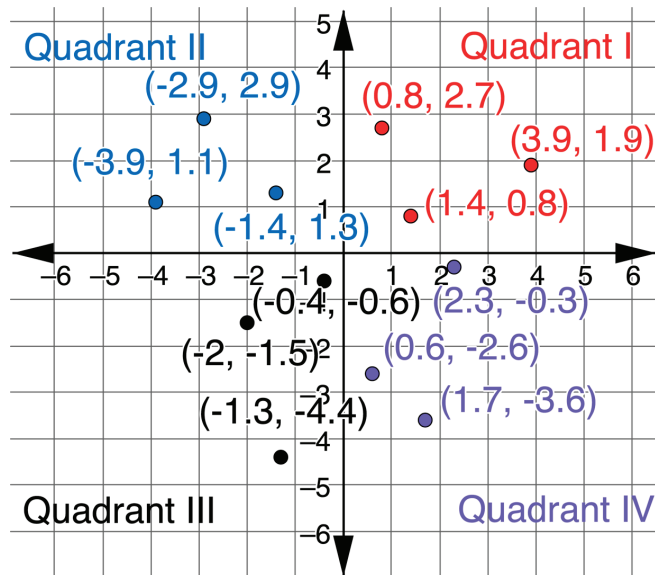
3. Four points are plotted on the coordinate grid.



a. What is the same about the points?

b. What is different about the points?

4. Twelve points are plotted on the coordinate grid. The grid shows the labels Quadrant I, Quadrant II, Quadrant III, and Quadrant IV.



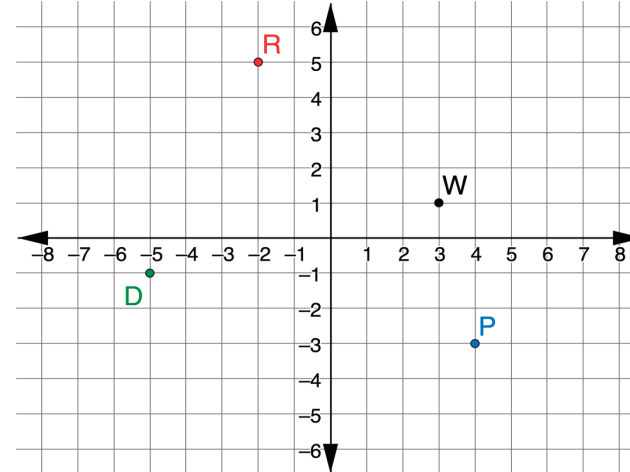
Work with your partner to summarize the characteristics of the points in each quadrant. Write your summary in the table.

Quadrant I	
Quadrant II	
Quadrant III	
Quadrant IV	



4.1.3 Bring It Together: Plotting Ordered Pairs

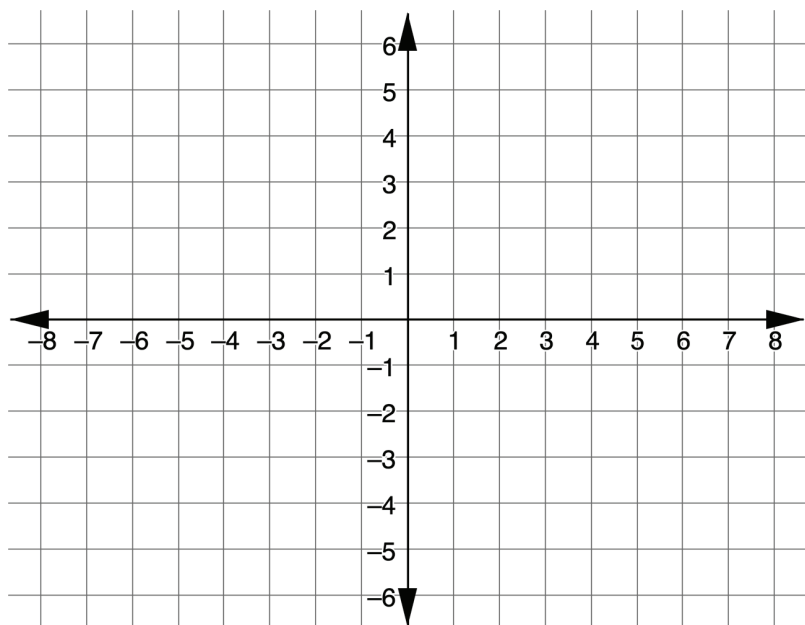
1. Points D , P , R , and W are plotted on the coordinate grid.



Select all of the descriptions of each point.

Point D	Point P
☐ The x -coordinate of D is -5 . ☐ The x -coordinate of D is -1 . ☐ The y -coordinate of D is -5 . ☐ The y -coordinate of D is -1 . ☐ Point D is in quadrant I. ☐ Point D is in quadrant II. ☐ Point D is in quadrant III. ☐ Point D is in quadrant IV.	☐ The x -coordinate of P is -3 . ☐ The x -coordinate of P is 4 . ☐ The y -coordinate of P is -3 . ☐ The y -coordinate of P is 4 . ☐ Point P is in quadrant I. ☐ Point P is in quadrant II. ☐ Point P is in quadrant III. ☐ Point P is in quadrant IV.
Point R	Point W
☐ The x -coordinate of R is -2 . ☐ The x -coordinate of R is 5 . ☐ The y -coordinate of R is -2 . ☐ The y -coordinate of R is 5 . ☐ Point R is in quadrant I. ☐ Point R is in quadrant II. ☐ Point R is in quadrant III. ☐ Point R is in quadrant IV.	☐ The x -coordinate of W is 1 . ☐ The x -coordinate of W is 3 . ☐ The y -coordinate of W is 1 . ☐ The y -coordinate of W is 3 . ☐ Point W is in quadrant I. ☐ Point W is in quadrant II. ☐ Point W is in quadrant III. ☐ Point W is in quadrant IV.

2. Use the coordinate grid to plot and label points that have the characteristics described in the table.



Point B has an x -coordinate of 0.	Point C has a y -coordinate of -4 and is in Quadrant III.
Point G has a y -coordinate of 0.	Point M has a y -coordinate of 2 and is in Quadrant I.
Point T has an x -coordinate of -3 and is in Quadrant II.	Point W has an x -coordinate of 1 and is in Quadrant IV.

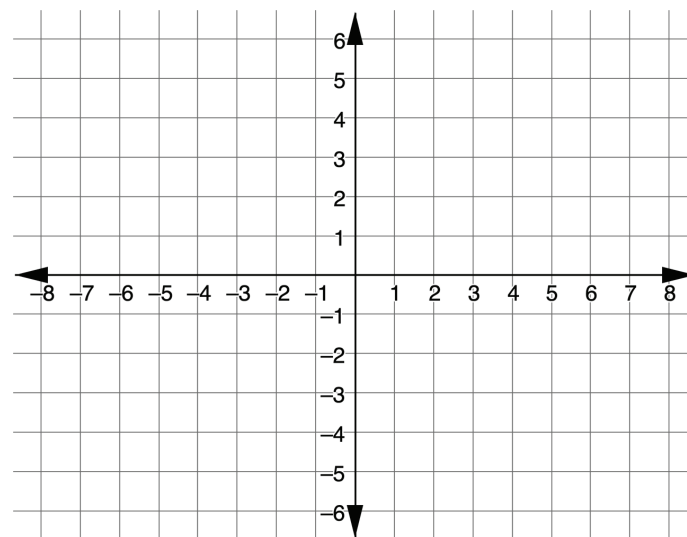


4.1.4 Guided Instruction: Reflections

1. Use the points in the table for the following.

Point D (1, 3)	Point H (1, -3)	Point K (-2 , 4)
Point R (-5 , -1)	Point V (5, -1)	Point Z (4, 4)

- a. Plot the ordered pairs from the table on the coordinate grid below. Label using the given letter.



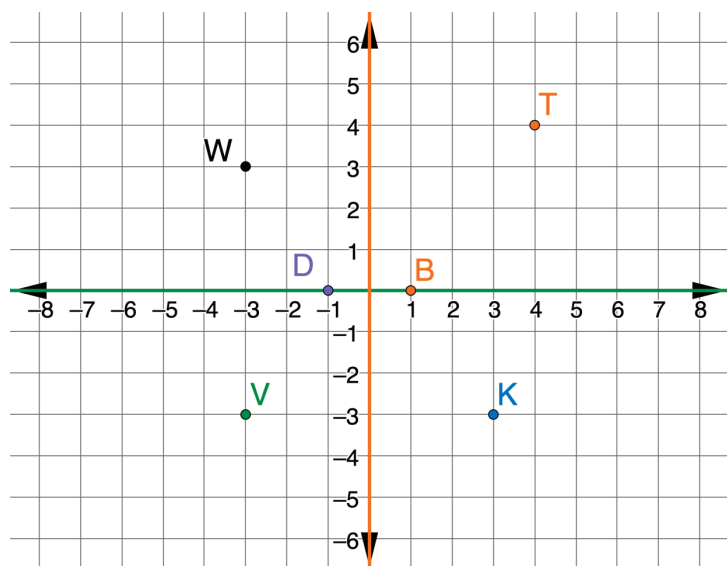
- b. Select the pair of points that have the same x -coordinate.

☐ D ☐ H ☐ K ☐ R ☐ V ☐ Z

- c. Select the pair of points that have a y -coordinate of -1 .

☐ D ☐ H ☐ K ☐ R ☐ V ☐ Z

2. A coordinate grid with points D , B , K , T , V , and W is shown. The x -axis is green and the y -axis is orange.

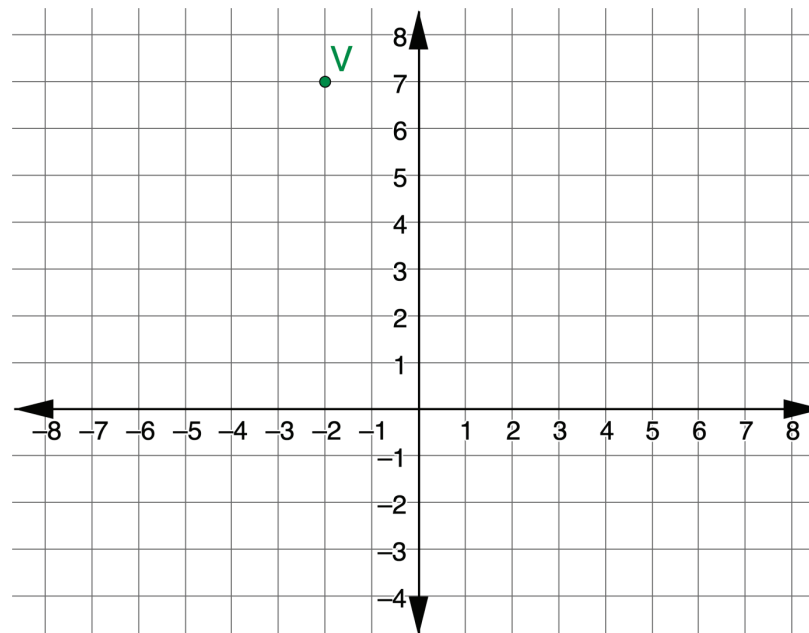


- Write the ordered pair of the reflection of Point V when the graph is folded across the x -axis.
- Write the ordered pair of the reflection of Point K when the graph is folded across the y -axis.
- Write the ordered pair of the reflection of Point D when the graph is folded across the y -axis.
- Write the ordered pair of the reflection of Point B when the graph is folded across the x -axis.



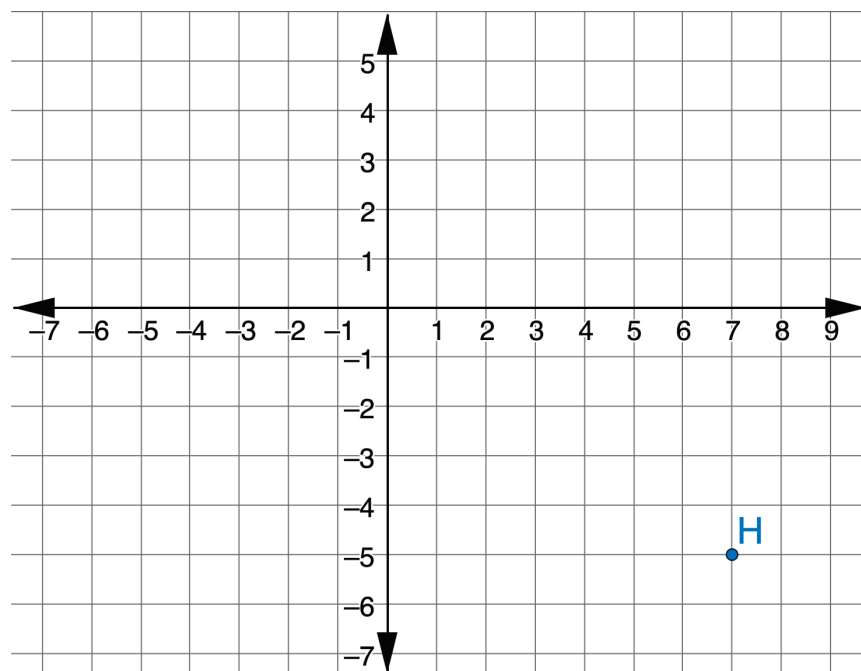
4.1.5 Guided Practice: Reflections

1. Point V is plotted on the coordinate grid.



What are the coordinates of the reflection of V across the y -axis?

2. Point H is plotted on the coordinate grid.

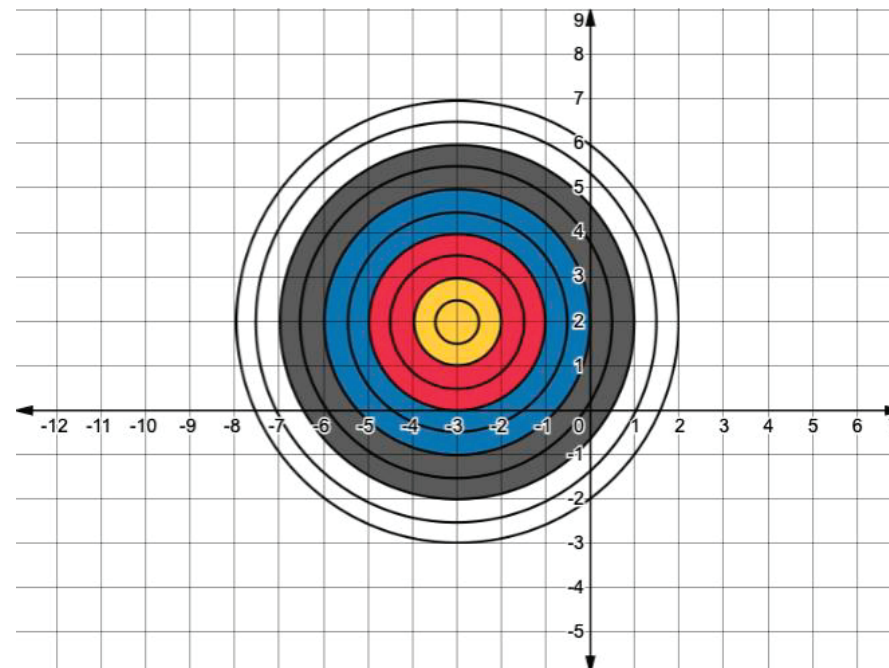


What are the coordinates of the reflection of H across the x -axis?



4.1.6 Wrap-Up: Ticket Out the Door

A target is shown on the coordinate grid.



1. Write the ordered pair that will hit the bullseye.
2. A player throws two darts. The first dart hits the ordered pair $(-2, -1)$. The second dart hits the point that is the reflection of $(-2, -1)$ over the x -axis. Complete the table that describes the location of the second dart.

Color	Ordered Pair

Unit 4, Lesson 2: Points on the Coordinate Plane – Part 1



Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y - coordinate.



Learning Targets

- ☐ I can plot integers using different scales.
- ☐ I can plot rational numbers in the form of decimal and fractional ordered pairs.
- ☐ I can select an appropriate scale given a set of ordered pairs.



4.2.1 Warm-Up: Interior Designer

After getting a degree in graphic design, Soledad Alzaga built her career and company as an interior designer. Interior designers make interior spaces functional, safe, and beautiful by determining space requirements and selecting decorative items such as colors, lighting, and materials. Beyond being creative and having good taste, Soledad says to own a successful interior design business, it is necessary to have the ability to work and communicate with others, be organized, have business knowledge, and to have math and measurement skills.

1. What is something that interests you that you would like to spend your career doing?

2. What skills do you think are needed to be successful in that career?



4.2.2 Refresher: Key Terms with the Coordinate Plane

Several key terms related to the coordinate plane were used throughout Lesson 1. Below, you can find definitions for each of these key terms.



The **coordinate plane** is a plane determined by two perpendicular number lines, called axes. The axes intersect at the origin. Each point in the coordinate plane is represented by a pair of coordinates that represent the direction and distance from each axis. The origin is represented by the coordinate pair, $(0, 0)$.



The **axes** of a graph are the horizontal and vertical number lines used in a coordinate plane system.



A **coordinate** is a number used to locate a point on a number line. A coordinate is one of the numbers in an ordered pair that locates a point on a coordinate plane.



The **origin** is the point of intersection of the x - and y -axes in a rectangular coordinate system, where the x -coordinate and y -coordinate are both 0.



A **quadrant** is any of the four regions separated by the axes in a coordinate plane.

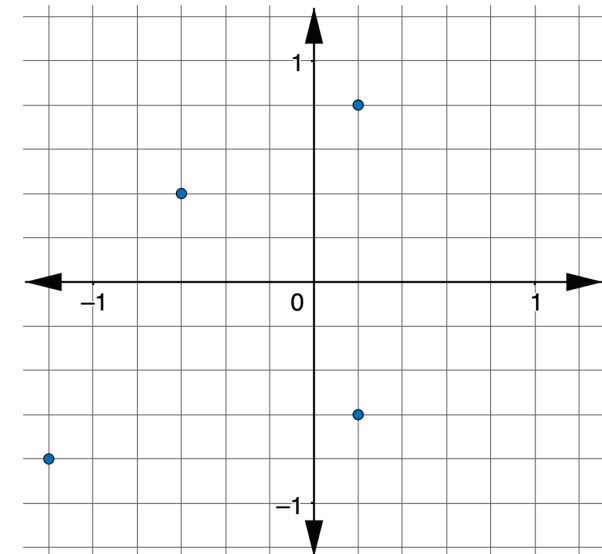


4.2.3 Guided Instruction: Plotting Points with a Fractional Scale

1. Four points are shown.

Point L	Point H	Point S	Point R
$\left(\frac{1}{5}, \frac{4}{5}\right)$	$\left(-\frac{3}{5}, \frac{2}{5}\right)$	$\left(-1\frac{1}{5}, -\frac{4}{5}\right)$	$\left(\frac{1}{5}, -\frac{3}{5}\right)$

- Based on the denominators of each fraction, what scale should be used for the grid?
- Find and label the points with the given letters on the coordinate plane.



2. Another set of four points is shown.

Point <i>M</i>	Point <i>L</i>	Point <i>C</i>	Point <i>A</i>
$(\frac{2}{3}, \frac{1}{6})$	$(\frac{1}{2}, -\frac{1}{3})$	$(-\frac{5}{6}, -\frac{1}{2})$	$(-\frac{1}{6}, 1)$

- a. List the denominators of each fraction.
- b. Zoe says she does not want to label her axes by halves because she will only be able to approximate the location of the thirds. She also says she does not want to label by thirds because there is no easy way to approximate the location of the halves. Christopher says she should label her axes by sixths.

Explain the benefit of using Christopher's plan.

- c. Draw and label coordinate axes with the scale you chose and plot the points.

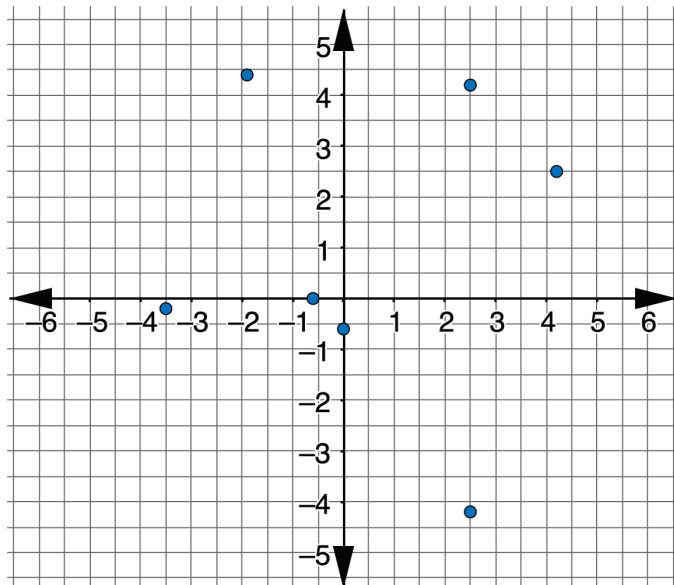


When deciding on a scale for plotting ordered pairs with rational values, consider the least common multiple (LCM) of each of the denominators to determine the proper increments.



4.2.4 Collaboration: Plotting Points with a Decimal Scale

1. A coordinate plane is shown.

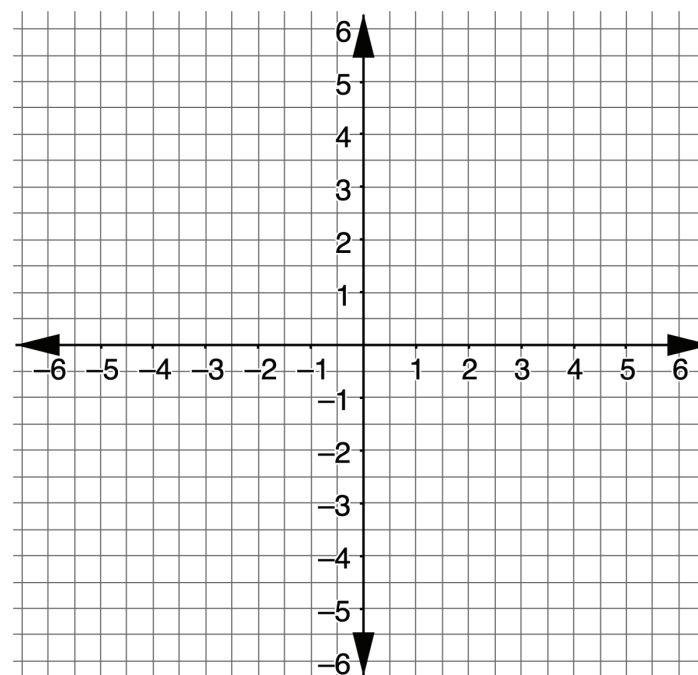


- a. Working with your partner, describe the scale that is being used for each axis.
- b. Locate each point on the coordinate plane and label it with the appropriate letter.

Point <i>G</i> (4.2, 2.5)	Point <i>B</i> (-1.9, 4.4)	Point <i>F</i> (0, -0.6)	Point <i>W</i> (-3.5, -0.2)
Point <i>E</i> (-0.6, 0)	Point <i>A</i> (2.5, 4.2)	Point <i>M</i> (2.5, -4.2)	

2. Working with your partner, plot and label the points using the given letters on the grid.

Point <i>L</i> (1.5, 2)	Point <i>B</i> (3.9, -4.5)	Point <i>G</i> (-5.5, -2.1)	Point <i>D</i> (0, 0.7)
Point <i>K</i> (-2.8, 5.7)	Point <i>S</i> (-0.8, 0)	Point <i>N</i> (-1, -3.4)	Point <i>H</i> (-3.9, 4.5)



- a. Which points were easy to plot?
- b. Which points were more difficult to plot?

3. On your own, answer the following question:

When plotting ordered pairs with decimal values, how is it best to determine the scale to use when setting up the axes on a grid?

My Reasoning

Share your reasoning with your partner and ask for feedback on any parts that were unclear or missing. Make any necessary edits to your reasoning.



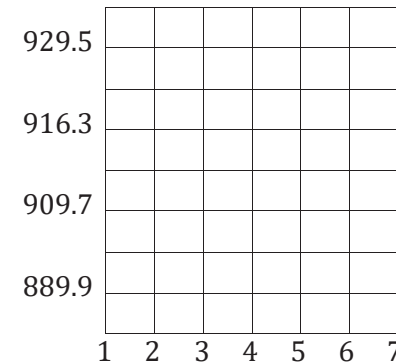
4.2.5 Collaboration: Choosing the Right Scale

When representing data, the scale of the grid is important to effectively and accurately communicate information.

- The table shows the total Florida tourism industry employees, in thousands, for years since 2002.

Years Since 2002	Total Tourism Industry Employees (in thousands)
1	889.9
4	909.7
5	916.3
7	929.5

To plot the data, Andrea drew the coordinate grid shown.

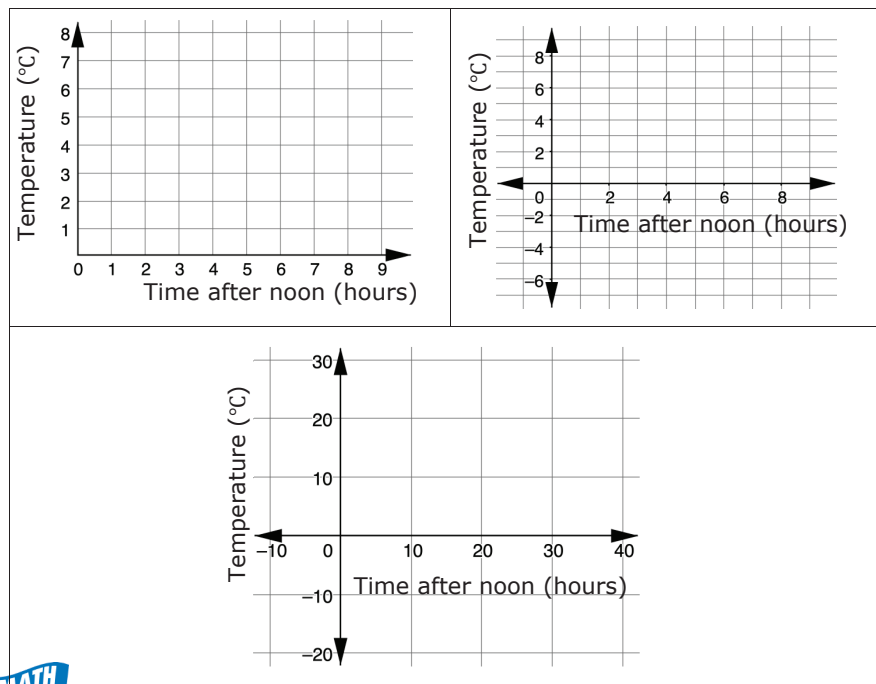


Discuss with your partner why Andrea's coordinate grid would not effectively nor accurately communicate the information about Florida tourism industry employees.

2. The data shown in the table are average autumn temperatures, in degrees Celsius ($^{\circ}\text{C}$), in Carmathen, Wales.

Time After Noon (hours)	Temperature ($^{\circ}\text{C}$)
0	5
1	4
2	3
3	2
4	1
5	-2
6	-3
7	-4
8	-4

Three coordinate grids are shown, each using different scales.

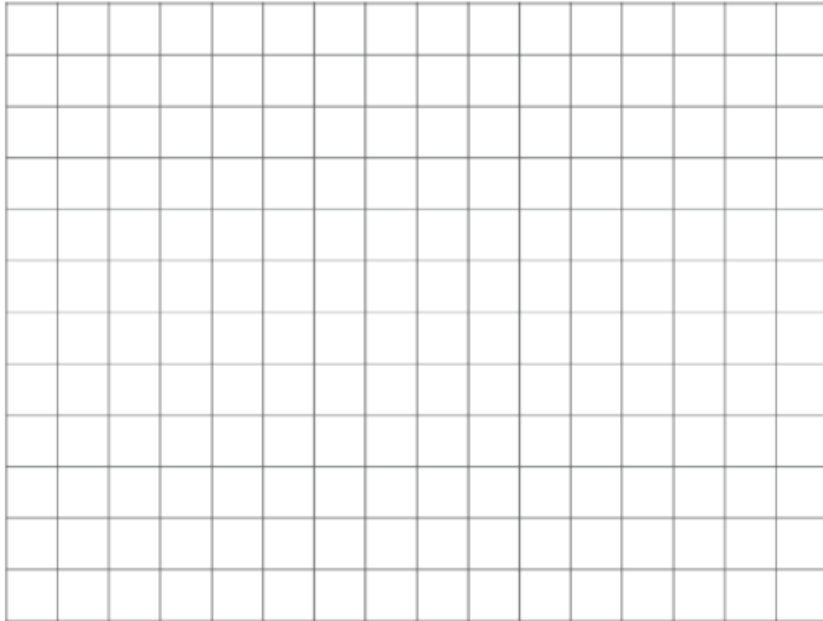


- a. Explain which coordinate grid most accurately represents the data.
- b. Explain why the other two coordinate grids may not be as accurate.



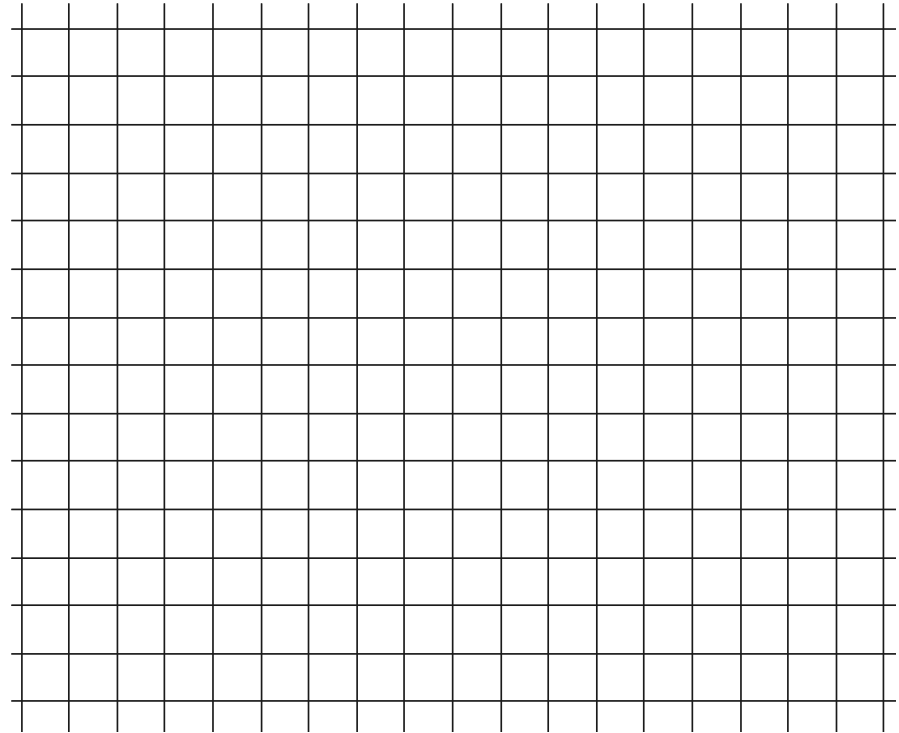
4.2.6 Try It: Plotting Points with Fractional and Decimal Scales

1. For the points $(1.75, -2.5)$ and $(-2.25, 1.5)$, draw and label coordinate axes with an appropriate scale, and plot the points.



2. Draw and label coordinate axes with an appropriate scale. Then, plot and label the points.

Point <i>W</i>	Point <i>D</i>	Point <i>P</i>	Point <i>T</i>
$\left(\frac{2}{3}, \frac{1}{4}\right)$	$\left(\frac{1}{2}, -\frac{1}{3}\right)$	$\left(-\frac{5}{3}, -\frac{1}{2}\right)$	$\left(-\frac{1}{6}, \frac{7}{4}\right)$





4.2.7 Wrap-Up: Reflection

1. Shade the arrow up to the statement that best describes your current level of understanding.



I'm so confident, I could explain plotting ordered pairs made up of fractions and/or decimals to someone else!

I can get to the right answer, but I don't understand it well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked the boxes I think I do not understand.

I tried hard and I listened, but I am finding this challenging. I checked the boxes I think I do not understand.

2. Check the skills that you currently understand.

- ☐ Determining the correct scale
- ☐ Identifying the LCM of the denominators for coordinates that include fractions
- ☐ Plotting ordered pairs that contain fractions
- ☐ Plotting ordered pairs that contain decimals



Unit 4, Lesson 3: Points on the Coordinate Plane – Part 2



Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.



Learning Target

- ☐ I can plot rational numbers using a scale of 1.



4.3.1 Warm-Up: Which One Is the Best?

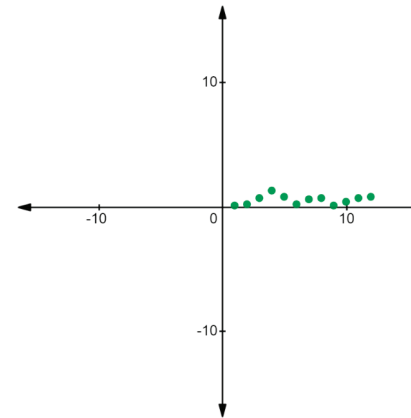
Coral polyps get stressed out and could die if the surrounding water is above 29°C . The table represents the average temperature, to the nearest tenth, above 29°C at the Great Coral Reef over a 12-week period.



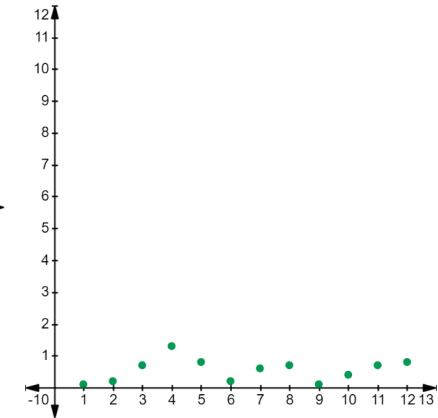
1. Explain which of the three graphs best represents the data. (Data taken from NOAA.)

Week	1	2	3	4	5	6	7	8	9	10	11	12
Temp	0.1	0.2	0.7	1.3	0.8	0.2	0.6	0.7	0.1	0.4	0.7	0.8

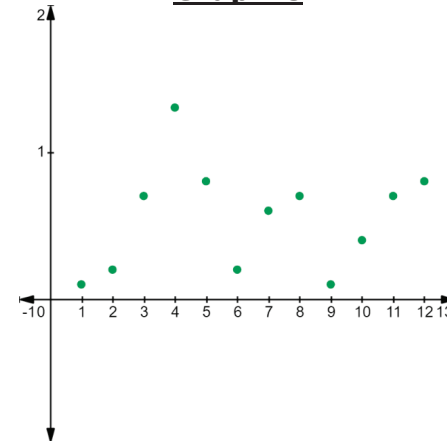
Graph A



Graph B



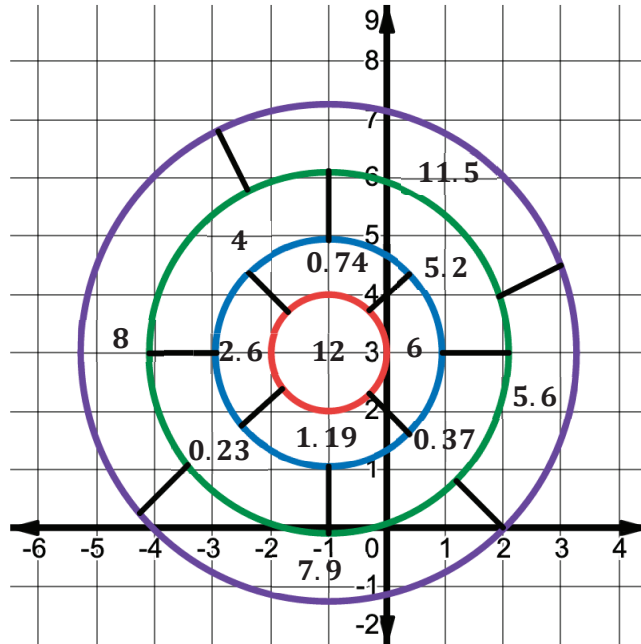
Graph C





4.3.2 Try It: Game Score

A gameboard is shown on the coordinate grid, along with the number of points earned for landing in each area. Each coordinate grid axis is scaled at 1.



1. Several points are given in the table. Locate each point on the coordinate grid. Complete the table by writing the score value for landing at each point. Calculate the game score by finding the sum of all the points. Verify your answer with your teacher.

Point	Score Value
$(-3, 4.9)$	
$(\frac{4}{5}, 1\frac{3}{4})$	
$(0, 2.2)$	
$(-3.8, 4.9)$	
$(1.5, 4.4)$	
$(-1\frac{2}{3}, 2)$	
$(\frac{5}{4}, \frac{11}{2})$	

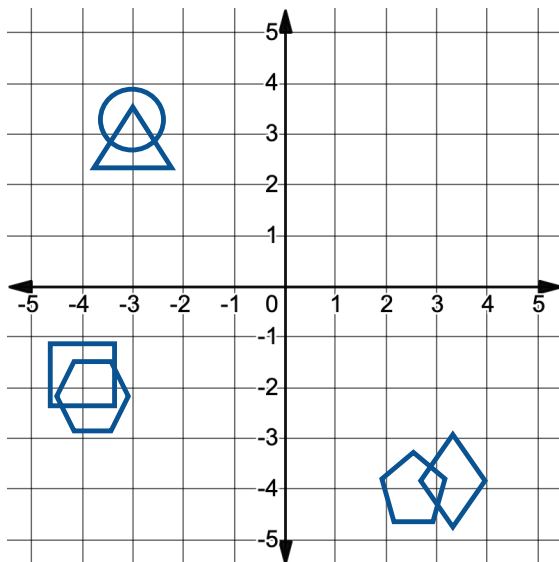
Game Score: _____



4.3.3 Collaboration: Hit or Miss

There are six shapes on the coordinate grid shown. Each partner will select one shape as their game piece. The object of the game is to guess which shape your partner selected before they guess yours. A hit means the point you guessed is inside their shape. A miss means that it is not. Keep your gameboard hidden from your partner during the game.

Gameboard



1. The shape of the region I am picking is a _____.

2. Record the coordinates of each guess in the table below. Then, plot the point on the coordinate grid gameboard.

Partner 1 Guess	Was it a hit or miss?	Partner 2 Guess	Was it a hit or miss?

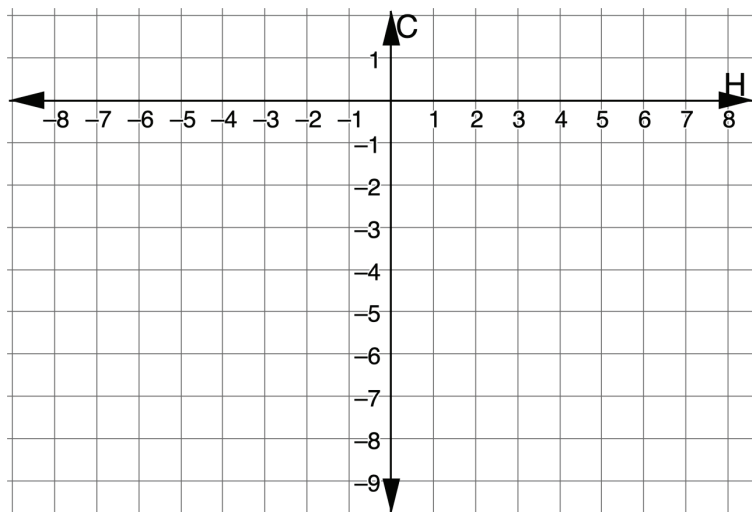


4.3.4 Your Turn!

1. The temperature in Princeton, New Jersey was recorded at various times during the day. The times relative to midnight, H , and temperatures in degrees Celsius ($^{\circ}C$) are shown in the table.

Time (hours before or after midnight, H)	Temperature ($^{\circ}C$)
-5	$\frac{6}{5}$
-2	$-1\frac{3}{5}$
0	$-\frac{7}{2}$
8	$-6\frac{7}{10}$

- a. Plot the points that represent the data.

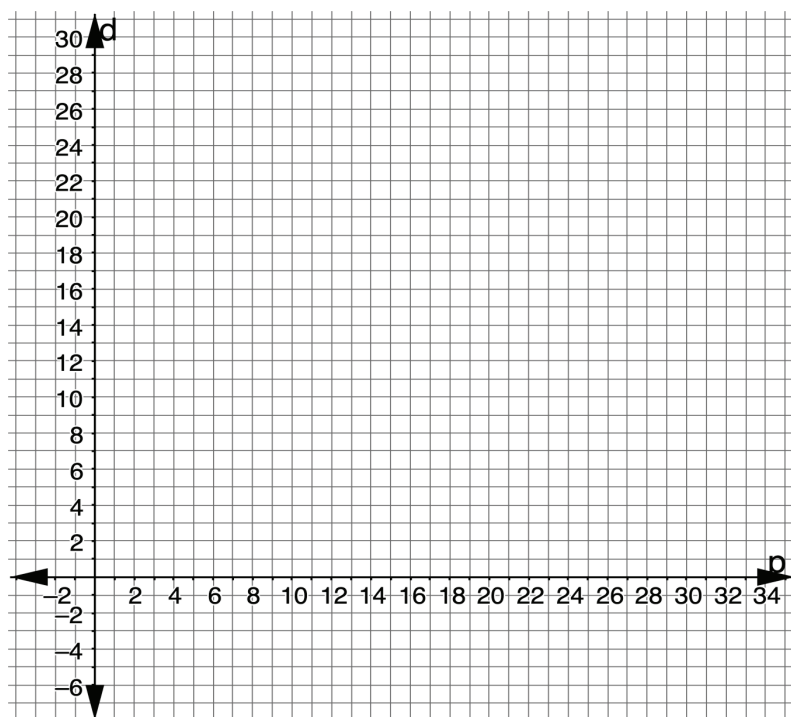


- b. In the town of New Haven, Connecticut, the temperature at midnight was $-\frac{2}{10}^{\circ}C$. Plot and label this point on the same grid. Label the point C.
- c. Label the point that represents the temperature in Princeton at midnight P . Which town was warmer at midnight, Princeton or New Haven?
- d. Explain how to determine which town you selected in Part C is warmer at midnight.

2. The average NBA player scores 9.7 points per game. The table below lists six players, along with their average points per game and their difference from the average.

Player	Average Points Per Game (p)	Difference from Average (d)
LeBron James	27.1	17.4
Dwyane Wade	22.0	12.3
Mike Conley	14.8	5.1
Frank Brickowski	10.0	0.3
Robin Lopez	8.8	-0.9
Greg Foster	3.9	-5.8

Graph the points from the table on the coordinate grid.



4.3.5 Wrap-Up: Reflection

Think about what you have learned in the past four lessons about plotting points. As you read each statement, consider which best describes your understanding right now.

1. I need help plotting points. I'm not sure where to begin.
 2. I can plot points with integers on a coordinate grid with a scale of 1, but I still need help with rational numbers and different scales.
 3. I can plot points with decimals on a coordinate grid with a tenths scale, but I still need help with points that are fractions.
 4. I can plot points with rational numbers that are decimals or fractions using any scale.
-
1. Explain which number from the list best describes you.
 2. Write a plan for what you can do to either move to number 4 or to help a classmate move to number 4.

Unit 4, Lesson 4: Distance on the Coordinate Plane



Benchmark

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.



Learning Target

- ☐ I can find the distance between two ordered pairs.



4.4.1 Warm-Up: A Point on the Number Line

- Circle all numbers that could be the location of point F on the number line.

3.5

$4\frac{1}{2}$

$-\frac{3}{5}$

-3.5

$-\frac{7}{2}$

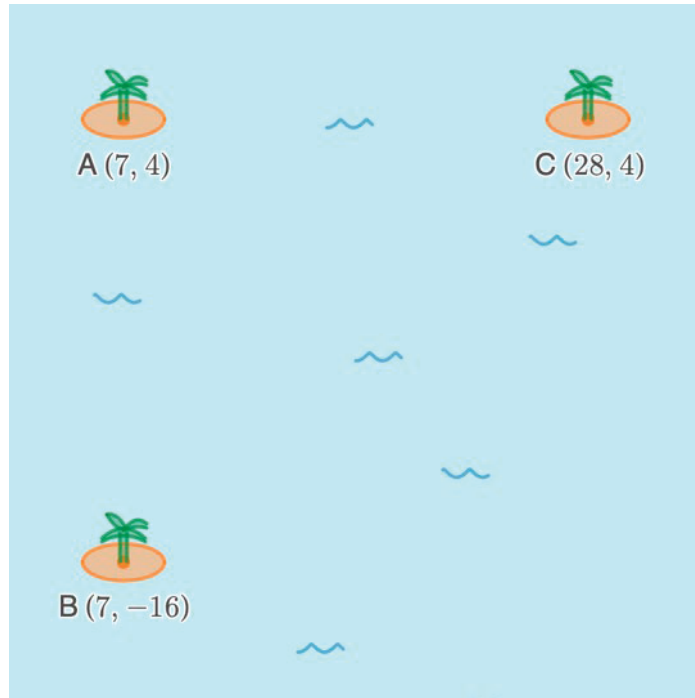
$-4\frac{1}{2}$





4.4.2 Collaboration: Who Is Correct?

The locations of three islands in the ocean are given using ordered pairs.



Three students described the distance between each island.

Braxton says the distance from point A to point C is **less than** the distance from point A to point B .

Justine disagrees with him and says the distance from point A to point C is **greater than** the distance from point A to point B .

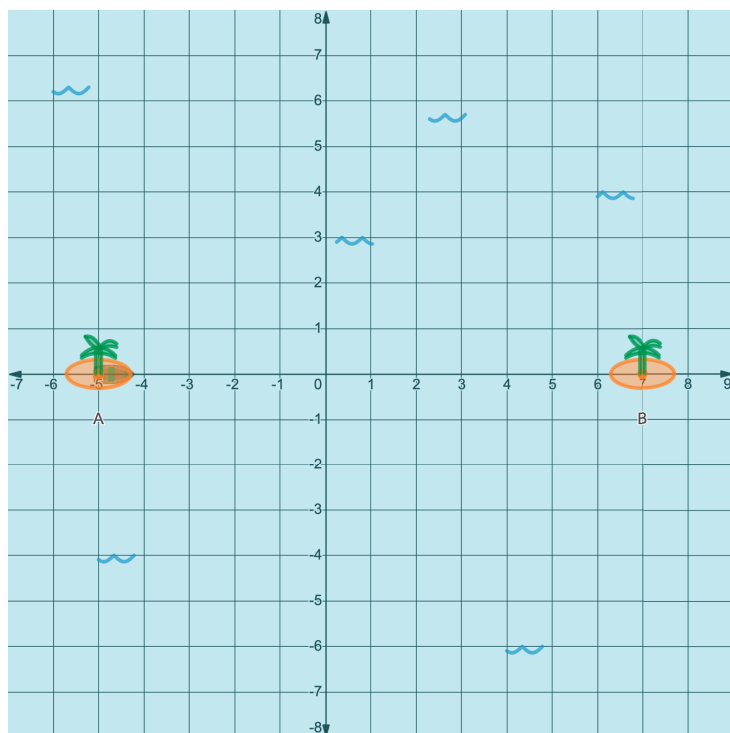
Rainey says the distance from point A to point C is **equal to** the distance from point A to point B .

1. Discuss with your partner who you think is correct. Write a summary of your discussion.



4.4.3 Guided Instruction: Distance on the Coordinate Plane

1. The coordinates of two small islands are labeled points A and B on the graph.



- a. Counting the grid lines, how far apart are the islands?

- b. What are the coordinates of points A and B ?

A : (____, ____)

B : (____, ____)

- c. Explain why the y -coordinates of the two points are the same.

- d. Describe the location of the two points relative to the y -axis.



The distance a number is from zero (0) on a number line is called the **absolute value**.

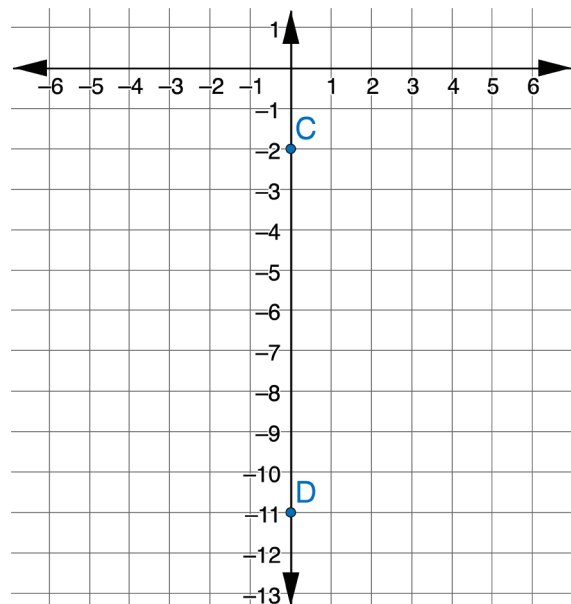
- e. What is the absolute value of the x -coordinate of each point?

The absolute value of the x -coordinate of Point A is ____.

The absolute value of the x -coordinate of Point B is ____.

- f. Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

2. Below is another coordinate plane with two different islands located at points C and D .



- How is the location of these two islands different from the first two you worked with?
- Counting the grid lines, how far apart are the islands?
- What are the coordinates of points C and D ?

C : (____, ____)

D : (____, ____)

- Explain why the x -coordinates of the two points are the same.

- Describe the location of the two points relative to the x -axis.

- What is the absolute value of the y -coordinate of each point?

The absolute value of Point C is _____.

The absolute value of Point D is _____.

- Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

3. Complete the statements.

When finding the horizontal distance between two

points that lie on the same line, if the points are on

opposite sides of the y -axis, ☐ add
☐ subtract the absolute
value of the unlike coordinates. If the points are on the

same side of the y -axis, then ☐ add
☐ subtract the absolute
values of the unlike coordinates.

When finding the vertical distance between two points

that lie on the same line, if the points are on the

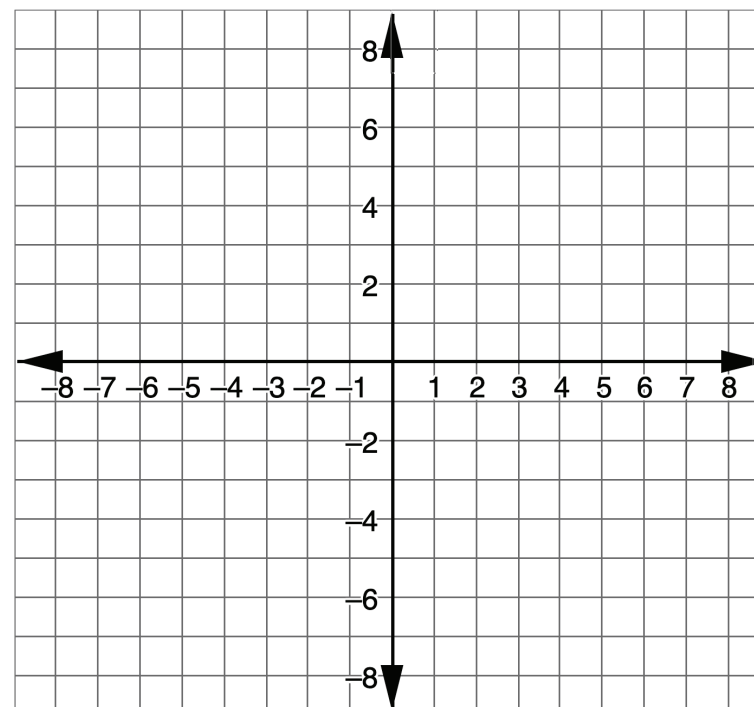
opposite sides of the x -axis, ☐ add
☐ subtract the absolute
value of the unlike coordinates. If the points are on the

same side of the x -axis, then ☐ add
☐ subtract the absolute
value of the unlike coordinates.



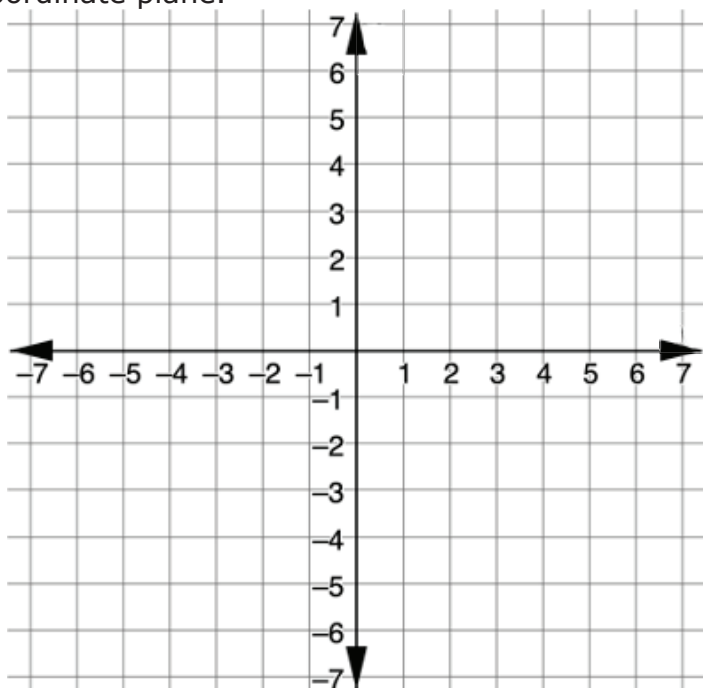
4.4.4 Guided Practice: Distances Between Rational Numbers

1. Graph the points $E(-5\frac{1}{3}, 4\frac{1}{2})$ and $F(7\frac{2}{3}, 4\frac{1}{2})$ on the coordinate plane.



- a. Describe the location of these points relative to the origin and the quadrants they lie in.
- b. What is the distance between point E and point F ?

2. Graph the points $K(-4, -3.25)$ and $M(-4, -0.5)$ on the coordinate plane.

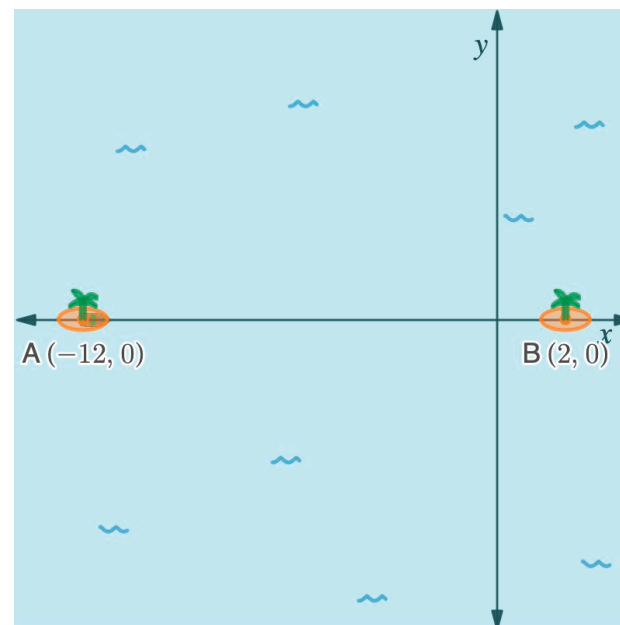


- a. Describe the location of these points relative to the origin and the quadrants they lie in.
- b. What is the distance between point K and point M ?

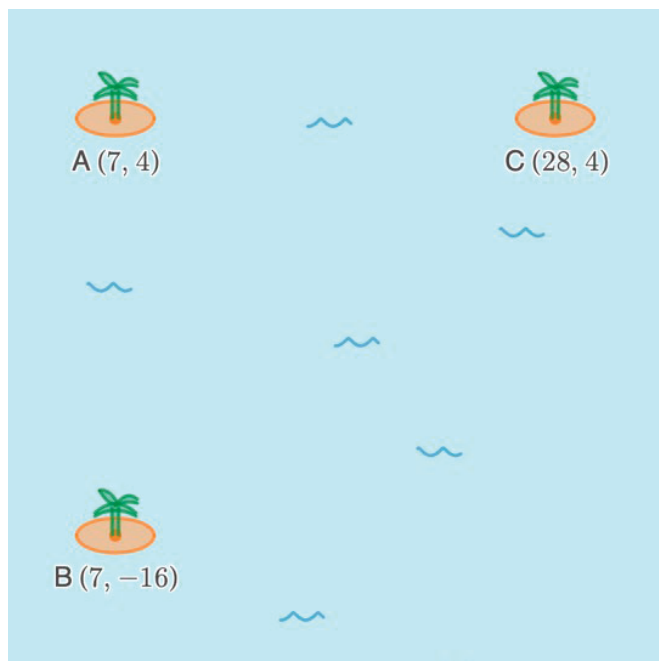


4.4.5 Your Turn!

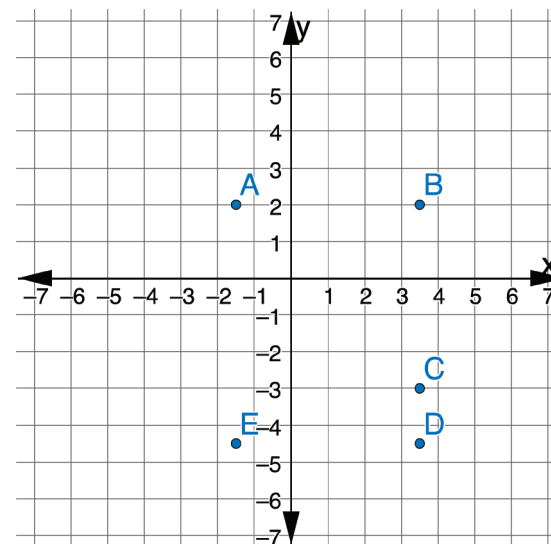
1. What is the distance from point A to point B ?



2. Explain which island is closest to island A . Show your work.



3. Five points are shown. Points A and E lie on the same vertical line. Points B, C , and D lie on a different vertical line. Points E and D lie on the same horizontal line.

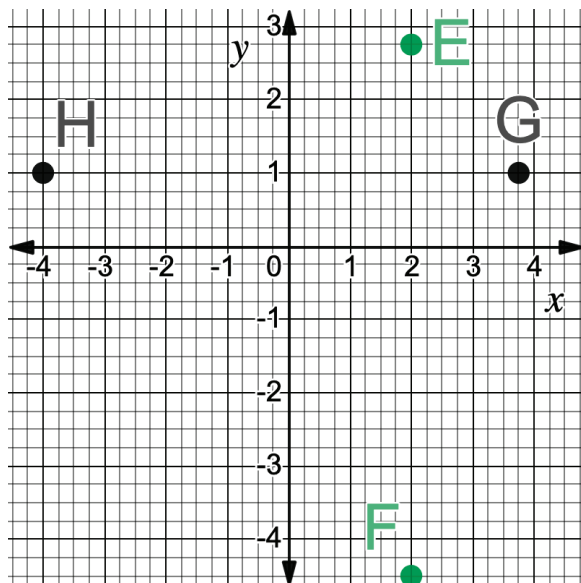


- a. Complete the table by filling in each missing coordinate.

A	$(-1.5, \underline{\hspace{1cm}})$
B	$(\underline{\hspace{1cm}}, 2)$
C	$(3.5, \underline{\hspace{1cm}})$
D	$(3.5, \underline{\hspace{1cm}})$
E	$(\underline{\hspace{1cm}}, -4.5)$

- b. The distance between points B and C is _____ units.
- c. The distance between points D and C is _____ units.
- d. The distance between points D and E is _____ units.

4. Use the coordinate plane below to determine distances between the following points.



- The distance between points E and F is _____ units.
- The distance between points G and H is _____ units.
- Which points are the same distance from $(2, 1)$?



4.4.6 Wrap-Up: Cool Down

- Your friend is absent today and will need the notes from class. Write a quick note to help them learn how to find the distance between two points on a coordinate plane.

Unit 4, Lesson 5: Perimeter on the Coordinate Plane



Benchmarks

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.



Learning Targets

- ☐ I can find distances between ordered pairs.
- ☐ I can find the dimensions of a rectangle on a coordinate plane.
- ☐ I can find the perimeter of a rectangle on a coordinate plane.



4.5.1 Warm-Up: Self-Assessment

1. Complete each statement to describe your understanding.

- ☐ I know
 - ☐ I need support with
- how to plot rational number

ordered pairs in all four quadrants.

- ☐ I know
 - ☐ I need support with
- how to find the perimeter and area

of rectangles.

- ☐ I know
 - ☐ I need support with
- how to find the vertical or

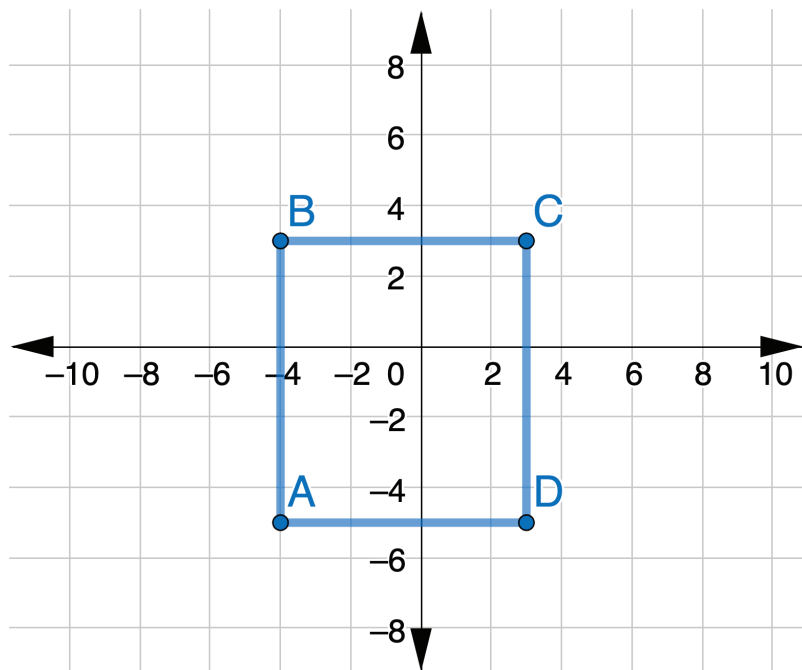
horizontal distance between two ordered pairs that are

located in the same or different quadrants.



4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles

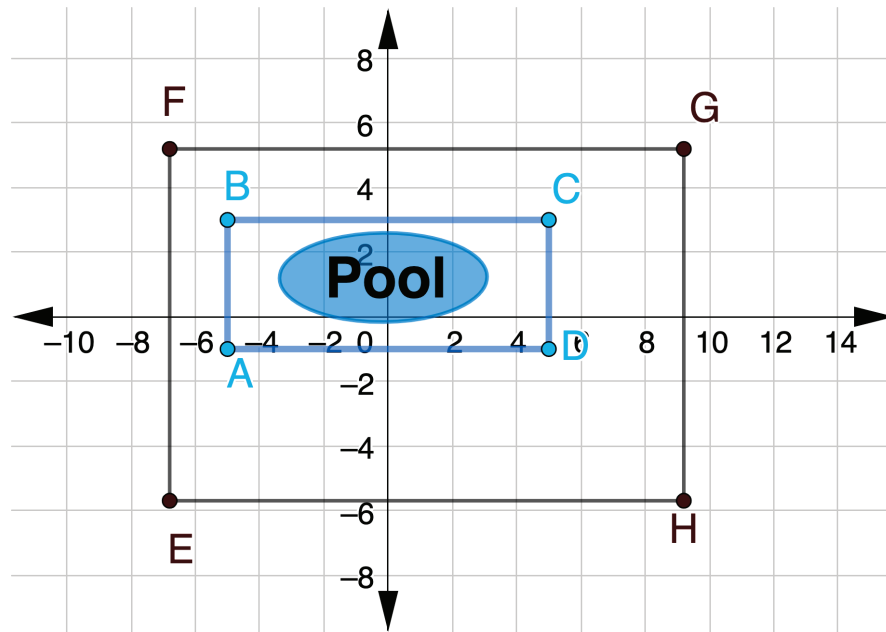
1. A rectangle has vertices at the points $A(-4, -5)$, $B(-4, 3)$, $C(3, 3)$, and $D(3, -5)$ as shown.



- a. The distance between points A and D is the length of the rectangle. The length of the rectangle is _____ units.

- b. What other two points on the rectangle are the same distance apart as points A and D ?
- c. Between which two points is the line segment that represents the width of the rectangle?
- d. Find the width of the rectangle.

2. Sonja's family installed a new pool. Sonja's dad wants to put up a fence, shown as a blue rectangle, around the pool for safety. The scale of the grid is represented in meters (m). He needs to know how much fencing he will need.



The rectangle that represents the fence has vertices at $A(-5, -1)$, $B(-5, 3)$, $C(5, 3)$, and $D(5, -1)$.

- What is the length of the line segment that connects points A and D ?
- What is the length of the line segment that connects points A and B ?

- Explain how to calculate the perimeter of the fence.

- What is the perimeter of the fence in meters?



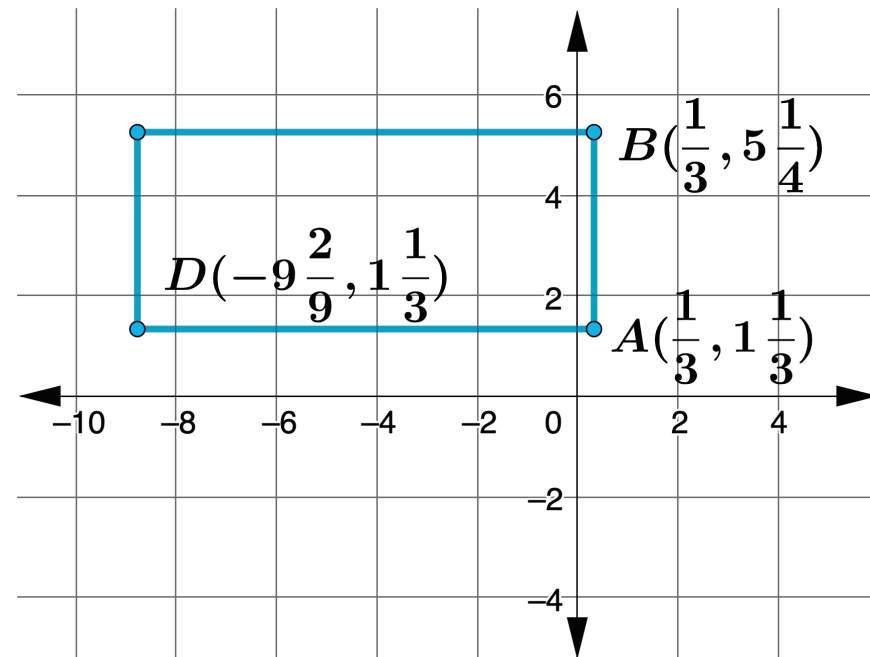
4.5.3 Try It: Perimeter on the Coordinate Plane

- Sonja's father plans to put another fence around the yard. The black rectangle on the coordinate grid in the previous component represents the second fence. The scale of the grid is represented in meters (m). The black rectangle has vertices at $E(-6.8, -5.7)$, $F(-6.8, 5.2)$, $G(9.2, 5.2)$, and $H(9.2, -5.7)$.
 - What is the length of the line segment that connects point E to point H ?
 - What is the length of line segment that connects point E to point F ?
 - Determine the perimeter of the fence around the yard.



4.5.4 Collaboration: Fix the Mistake

- Vera and Conor are working on their homework together to find the perimeter of the rectangle, shown on the coordinate plane. The width of the rectangle is the segment between point A and point B .



Conor counts the units between the numbers, while Vera determines the dimensions by subtracting. Each student's work and answers are shown.

Vera's Thinking	Conor's Thinking
Vera finds the dimension of the width by subtracting the y-coordinates. $5\frac{1}{4} - 1\frac{1}{3} =$ $\frac{21}{4} - \frac{4}{3} =$ $\frac{63}{12} - \frac{16}{12} =$ $\frac{47}{12}$ Vera says the width of the rectangle is $3\frac{11}{12}$ units.	Conor notices there is 1 full unit between point <i>A</i> and point <i>B</i> and two pieces that look like they would combine to make another whole unit. Conor says the width of the rectangle is 2 units.

Discuss each student's work with your partner. Explain any errors that either student made.

-
2. The length of the rectangle is the segment between point A and point D .

Vera finds the dimensions of the length by subtracting the x -coordinates.

$$9\frac{2}{9} - \frac{1}{3} =$$

$$\frac{83}{9} - \frac{3}{9} =$$

$$\frac{80}{9}$$

Vera finds the length of the rectangle to be $8\frac{8}{9}$ units.

Conor counts the units between the numbers again. He counts 4 units between A and D , but then realizes each space on the grid represents 2 units, so he doubles this to 8. He then determines $\frac{2}{9}$ of a unit plus $\frac{1}{3}$ of another unit equals $\frac{5}{9}$. He adds this to the 8.

Conor finds the length of the rectangle to be $8\frac{5}{9}$ units.

Discuss each student's work with your partner. Explain any errors that either student made.

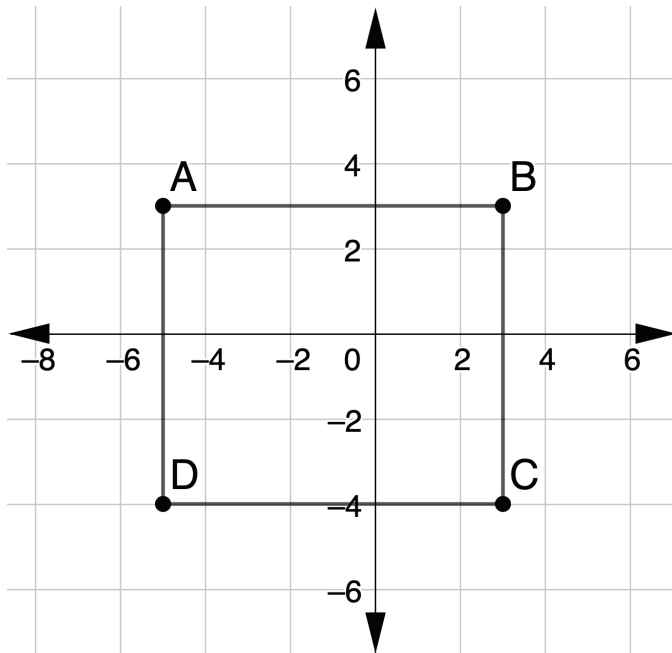
3. After collaborating with each other and understanding their mistakes, Vera and Conor each correctly find the perimeter of the rectangle. What perimeter did they find?





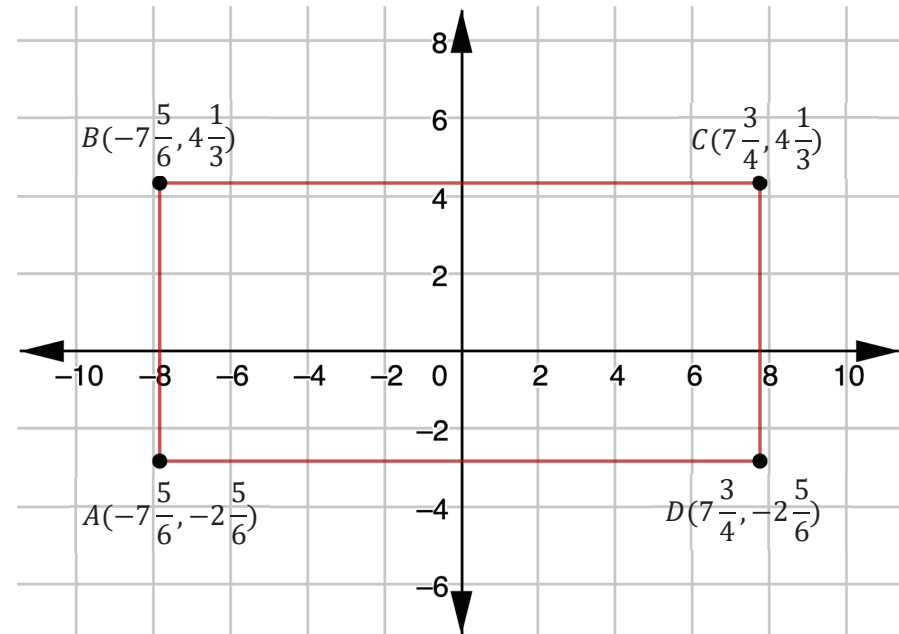
4.5.5 Your Turn!

1. The vertices of a rectangle are at points $A(-5, 3)$, $B(3, 3)$, $C(3, -4)$, and $D(-5, -4)$.



- What is the length, in units, of the rectangle?
- What is the width, in units, of the rectangle?
- What is the perimeter, in units, of the rectangle?

2. Titus wants to put a border around the outside of his garden.

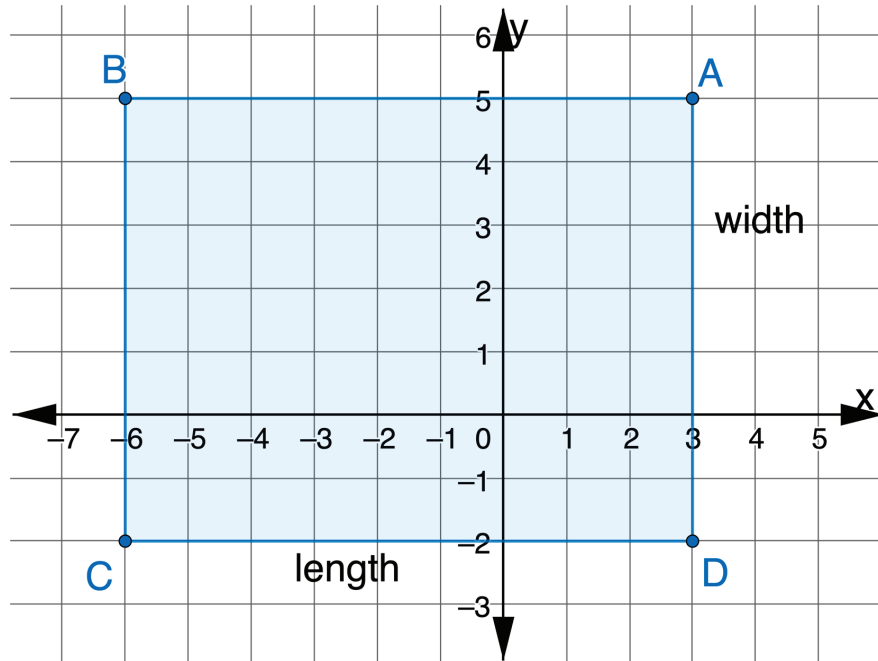


If the scale of the grid is in feet (ft), how many feet of border will he need to buy to completely fence in the garden?



4.5.6 Wrap-Up: Ticket Out the Door

1. A rectangle is shown on the coordinate plane.



- What is the length, in units, of the rectangle?
- What is the width, in units, of the rectangle?
- What is the perimeter, in units, of the rectangle?



Unit 4, Lesson 6: Solving Problems Involving Rectangles on the Coordinate Plane



Benchmarks

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.



Learning Targets

- ☐ I can find distances between ordered pairs.
- ☐ I can find the area of a rectangle on a coordinate plane.
- ☐ I can find the fourth vertex of a rectangle on a coordinate plane.



4.6.1 Warm-Up: True Statement

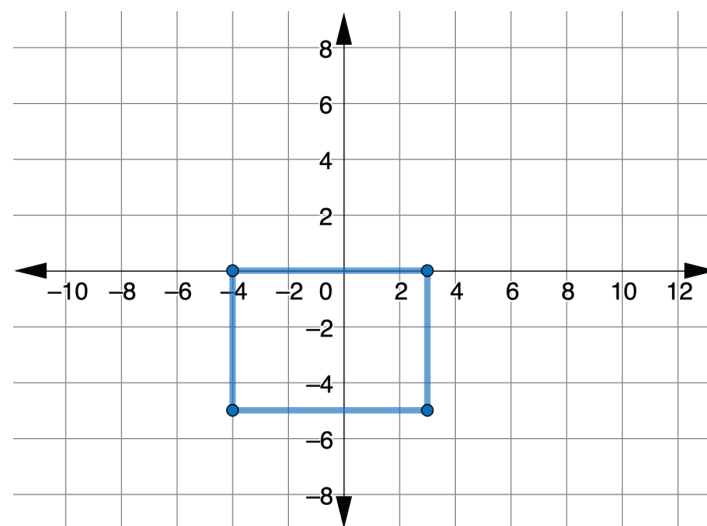
1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\boxed{}.\boxed{} \times 3.2 = \boxed{}.\boxed{}$$



4.6.2 Collaboration: Area of Rectangles

1. Work with your partner to explain how to find the area of a rectangle.
2. A rectangle with coordinate $(3, 0)$, $(3, -5)$, $(-4, 0)$, and $(-4, -5)$ is shown. The scale of the grid is in centimeters (cm).



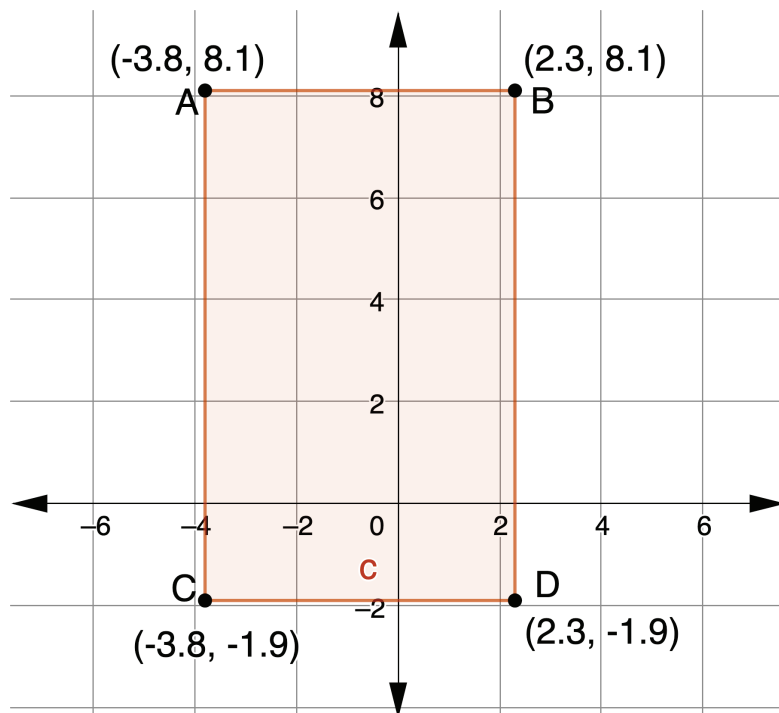
Work with your partner to answer the following questions.

- a. Determine the length and the width of the rectangle. Include units.
- b. What is the area of the rectangle? Include units.



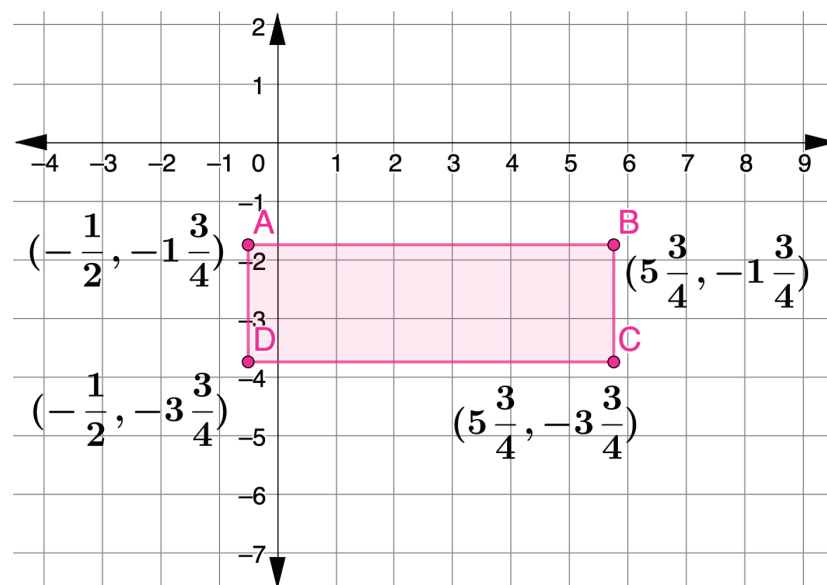
4.6.3 Guided Practice: Finding the Area of Rectangles on the Coordinate Plane

1. Henry wants to install new carpet in his living room. A drawing of his living room, with measurements in feet, is shown.



What is the area of the living room?

2. A rectangle is shown.



What is the area of the rectangle?



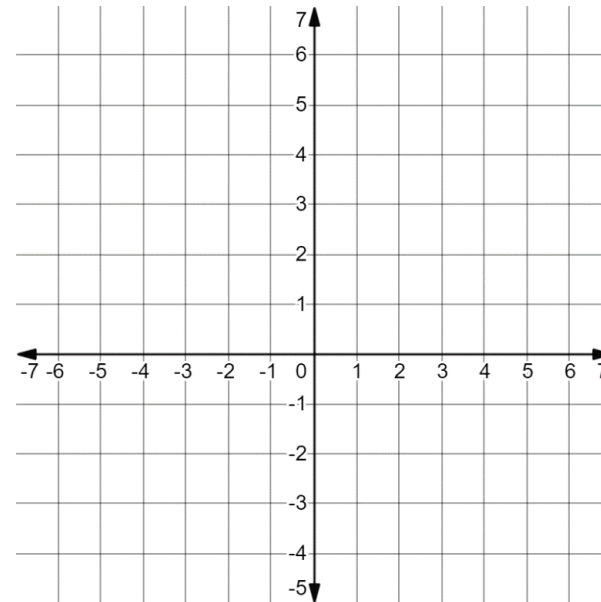
4.6.4 Collaboration: Finding the Fourth Vertex

1. With your partner, list every attribute you know about rectangles.

Rectangles

2. With your partner, find another pair of partners. Share what you wrote about rectangles. Add any new attributes the other group found that you didn't.

3. On the coordinate grid, draw a rectangle whose sides lie on the gridlines. Label the ordered pair of each vertex.

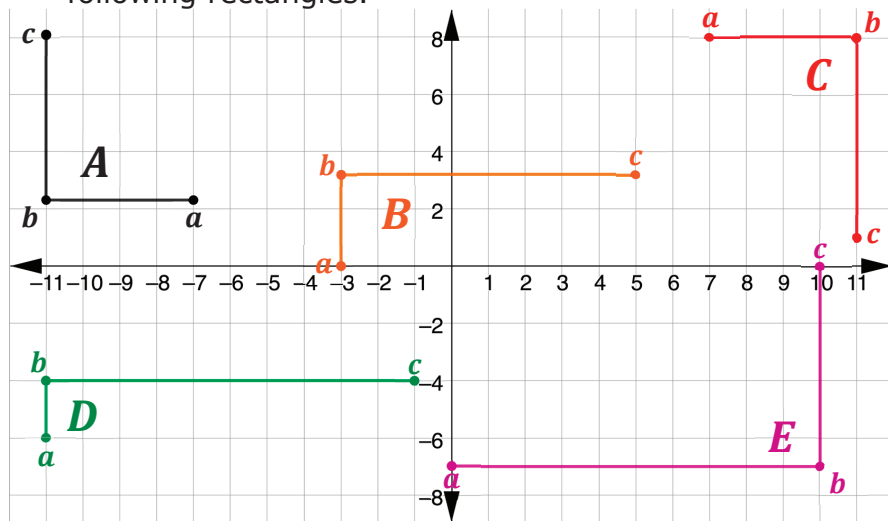


4. Discuss with your partner how to determine where to place the vertices of your rectangle. Write a summary of your discussion.
5. Based on the rectangle drawn in the previous questions, add any attributes about rectangles to your list.



4.6.5 Guided Practice: Finding the Fourth Vertex

1. Using the coordinate plane, find the fourth vertex of the following rectangles.



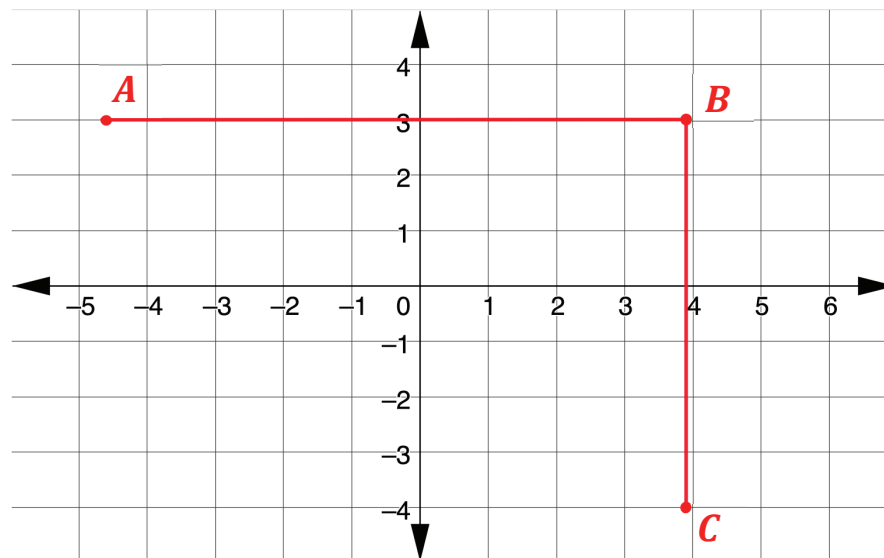
	Vertex a	Vertex b	Vertex c	Vertex d
Rectangle A	$(-7, 2.3)$	$(-11, 2.3)$	$(-11, 8.1)$	
Rectangle B	$(-3, 0.2)$	$(-3, 3.2)$	$(5, 3.2)$	
Rectangle C	$(7, 8)$	$(11, 8)$	$(11, 1)$	
Rectangle D	$(-11, -6)$	$(-11, -4.1)$	$(-1, -4.1)$	
Rectangle E	$(0, -7)$	$(10, -7)$	$(10, 0)$	

2. What is the area of Rectangle A, in square units?



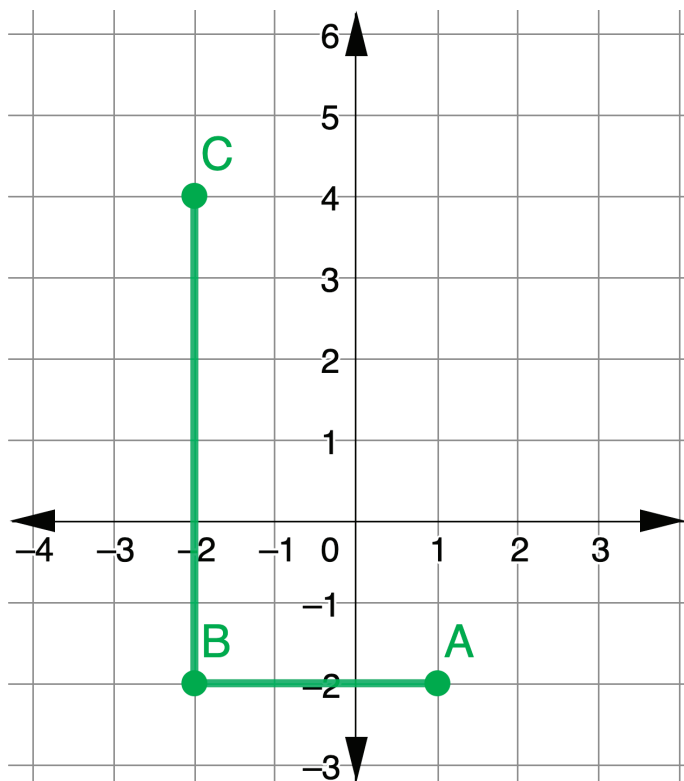
4.6.6 Your Turn!

1. Three vertices of a rectangle are at points $A(-4\frac{3}{5}, 3)$, $B(3\frac{9}{10}, 3)$, and $C(3\frac{9}{10}, -4)$. Each gridline represents 1 centimeter.



- What are the coordinates of the fourth vertex?
- What is the area of the rectangle?

2. Jada needs to cut out a piece of fabric in the shape of a rectangle. She laid out the fabric on a cutting board, whose scale is represented in feet (ft). Jada placed two edges of the fabric at points A and B . She wants the length of the fabric to be as long as the distance from point B to point C .



- Determine the coordinates of the fourth corner of the fabric.
- Find the area of the piece of fabric.



4.6.7 Wrap-Up: Reflection

- Complete the following statements.
 - Today I am still unsure about ...
 - To fill this gap, I intend to ...